Total Artificial Heart Emergency

AC-15 | Adult Cardiac

Quick flow

01 Identify Total Artificial Heart [EMT / AEMT / Paramedic]

Confirm TAH/Freedom or Companion driver • gather trained caregiver and backup equipment

02 Assess Perfusion [EMT / AEMT / Paramedic]

Mental status • ventilation • skin • capillary refill • palpable pulse • BP • $EtCO_2$ when available

03 Contact Transplant Center [EMT / AEMT / Paramedic]

Call coordinator immediately • SynCardia clinical emergency support 866-771-9437 • do not delay treatment

04 Pulse / Perfusion Absent? [EMT / AEMT / Paramedic]

Check both air drivelines, kinks, connections, driver power, battery charge, and backup driver

05 Low BP / Poor Perfusion [AEMT / Paramedic]

SBP <90: isotonic fluid 500 mL IV/IO • may repeat once • maximum 1 L

06 High BP + Pulmonary Edema [AEMT / Paramedic]

SBP ≥150: oxygen/CPAP • nitroglycerin 0.4 mg SL every 5 min • maintain SBP ≥90

07 Output Not Restored [EMT / AEMT / Paramedic / Medical Control]

No CPR/defibrillation/pacing/ACLS drugs • coordinator + Medical Control • death-in-field policy

08 Specialty Transport [EMT / AEMT / Paramedic]

TAH-capable/implanting center • caregiver • primary and backup drivers • batteries • chargers • records

Overview

- Device-specific emergency pathway for a patient supported by a SynCardia Total Artificial Heart (TAH). This is not an LVAD; use AC-14 for a durable LVAD.
- A TAH replaces both native ventricles and all four native valves. The external pneumatic driver produces pulsatile circulation, so a functioning system should generally produce a palpable pulse and measurable blood pressure.
- Standard chest compressions, ECG-based rhythm treatment, defibrillation, cardioversion, pacing, antiarrhythmics, and vasopressors do not restore TAH output and must not be used for TAH pump failure.

Provider-Level Actions

Restore pneumatic driver output, support ventilation and perfusion, and avoid standard cardiac-arrest interventions that cannot operate a TAH.

EMT

- Confirm the device is a TAH rather than an LVAD; assess ventilation, pulse, blood pressure, skin perfusion, mental status, and driver function.
- Provide airway and ventilatory support, expose the driver and both air drivelines, correct an obvious kink or loose external connection, and confirm power and battery charge.
- Contact the transplant coordinator, keep the trained caregiver with the patient, gather all primary and backup equipment, and do not initiate CPR or AED use for TAH pump failure.

AEMT

- Perform all EMT actions plus IV/IO access and isotonic fluid 500 mL for SBP below 90 mmHg or poor perfusion; may repeat once to a maximum of 1 liter.

- For SBP at least 150 mmHg with pulmonary edema or device-related symptoms, support oxygenation/CPAP and give nitroglycerin 0.4 mg SL every five minutes while maintaining SBP at least 90 mmHg.
- Assist a trained caregiver or coordinator-directed driver exchange without disconnecting both pumping lines simultaneously.

Paramedic

- Perform all AEMT actions plus advanced airway and waveform-capnography support when indicated; trend blood pressure, pulse, perfusion, and $EtCO_2$ rather than ECG rhythm.
- Coordinate troubleshooting, Medical Control, transplant-center consultation, specialty destination, and the death-in-field decision when output cannot be restored.
- Do not give vasopressors or antiarrhythmics and do not perform defibrillation, cardioversion, pacing, or chest compressions for TAH pump failure.

Indications

- Patient supported by a SynCardia Total Artificial Heart with illness, injury, poor perfusion, abnormal blood pressure, driver alarm, loss of pulse, or suspected system failure.

Assessment | Patient and perfusion assessment

- Assess responsiveness, airway, ventilation, skin color and temperature, capillary refill, palpable peripheral pulse, manual or automated blood pressure, and $EtCO_2$ when available.
- A functioning TAH produces pulsatile flow. Absent pulse or blood pressure indicates absent or severely impaired artificial-heart output until proven otherwise.
- Do not use ECG or 12-lead findings to determine TAH output. The native ventricles are absent, and the electrical tracing does not drive or represent artificial-heart circulation.
- Look for hypovolemia, hemorrhage, pulmonary edema, stroke, infection or sepsis, respiratory failure, trauma, driver failure, power loss, and air-driveline obstruction.

Assessment | Driver and driveline assessment

- Expose the external driver, power source, batteries, and both right- and left-sided pneumatic drivelines.
- Ensure each air driveline is firmly connected and is not kinked, bent, compressed, cracked, or disconnected.
- Confirm the driver is operating, read available alarms, verify external power connections, and check each battery charge indicator.
- Use the patient's backup driver only with a trained caregiver or direct transplant-coordinator guidance whenever possible.

Treatment steps

- Support airway and ventilation and contact the transplant coordinator immediately without delaying device assessment or treatment.
- For absent pulse or poor perfusion, correct external air-driveline kinks or loose connections, restore power, replace depleted batteries as directed, and transition to the backup driver when available with trained assistance.
- Do not disassemble the driver, force damaged connections, or disconnect both pumping lines simultaneously.
- For SBP below 90 mmHg or poor perfusion after confirming driver function, give isotonic crystalloid 500 mL IV/IO; may repeat once to a maximum of 1 liter with reassessment.
- For SBP at least 150 mmHg with pulmonary edema or device-related symptoms, provide oxygen/CPAP and nitroglycerin 0.4 mg SL every five minutes while maintaining SBP at least 90 mmHg.
- Do not give furosemide or administer the patient's home antihypertensive under this protocol.
- Treat hemorrhage, stroke, infection, respiratory failure, trauma, or another concurrent emergency under the linked complaint-specific protocol without disrupting the TAH system.

- If pulse and perfusion cannot be restored after troubleshooting, backup-driver exchange, ventilation, and appropriate fluid, contact the transplant center and Medical Control and follow the death-in-field policy.
- Transport to the implanting or TAH-capable transplant center when feasible with the caregiver and all primary and backup equipment.

Medications

Isotonic crystalloid

500 mL IV/IO; may repeat once to maximum 1 L

- For SBP below 90 mmHg or poor perfusion after confirming driver function.
- Reassess pulse, blood pressure, skin perfusion, lung sounds, ventilation, and $EtCO_2$ after each bolus.

Nitroglycerin

0.4 mg SL every 5 minutes

- For SBP at least 150 mmHg with pulmonary edema or device-related symptoms.
- Maintain SBP at least 90 mmHg.

Warnings / precautions

- Do not perform manual or mechanical chest compressions on a TAH patient.
- Do not use ECG rhythm, 12-lead ECG, AED, defibrillation, synchronized cardioversion, or transcutaneous pacing to treat TAH pump failure.
- Do not administer antiarrhythmics or vasopressors for loss of TAH output.
- Do not disconnect both pneumatic drivelines simultaneously, disassemble the external driver, or force a damaged connection.
- Do not confuse a Total Artificial Heart with an LVAD; the AC-14 perfusion and CPR pathway does not apply to a TAH.
- Do not add furosemide or assist with the patient's home antihypertensive under this protocol.

Clinical pearls

- The TAH replaces both ventricles and all four native valves while preserving the atria and great-vessel roots.
- The right and left artificial ventricles are powered pneumatically through two separate air drivelines; kinking either line can stop or reduce output.
- The device produces pulsatile flow, so a functioning TAH usually produces a palpable pulse and measurable blood pressure.
- $EtCO_2$ can help trend perfusion when ventilation is controlled, but restoration of driver output is the essential intervention.
- The trained caregiver and transplant coordinator are critical clinical resources and may be the most experienced people present with the patient's exact equipment.

Special populations | Anticoagulation, bleeding, and stroke

- Expect anticoagulant or antiplatelet therapy until verified and control external hemorrhage immediately.
- Maintain a low threshold for gastrointestinal bleeding, intracranial hemorrhage, thromboembolic stroke, and device-associated infection.

Special populations | DNR, MOST, or obvious death

- Honor a valid directive under agency policy and use the separate obvious-death or death-in-field policy when applicable.
- When pump output cannot be restored and no effective circulatory intervention remains, contact Medical Control and the transplant center before disposition whenever feasible.

Related pathways / contacts

Call SynCardia Clinical Emergency Support - 24-hour clinical support for qualified healthcare providers: 866-771-9437.

AC-14 LVAD Emergency - Use instead when the patient has a durable left ventricular assist device rather than a total artificial heart.

AR-01 Adult Airway - Open for airway or ventilatory failure.

AR-04 COPD / Asthma - Open for respiratory distress or bronchospasm when appropriate.

UP-04 Altered Mental Status - Open for persistent altered mental status or a concurrent metabolic cause.

UP-14 Stroke - Open for focal neurologic deficit or suspected stroke.

UP-15 Sepsis - Open for suspected infection, sepsis, or shock.

AM-05 Hypotension / Shock - Open for concurrent non-device shock after driver function is restored.

Source and references

SynCardia Systems. SynCardia Total Artificial Heart and Freedom Driver clinical information and emergency support guidance, current edition. | North Carolina College of Emergency Physicians. AC-15 Total Artificial Heart Protocol, revised September 1, 2025. | Claiborne EMS airway, stroke, sepsis, shock, bleeding, and death-in-field protocols.

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